



PROJECT

TWT2231 – WEB TECHNIQUES AND APPLICATIONS

TRIMESTER: Mar/APR 2026 (Term 2610)

Project Title: Employee Leave Management System

Lab Section: 1F

Submission Date: xx XXX 2026

NO	STUDENT ID	STUDENT NAME	MAJOR
1.	242UT244D2	YEAP BOON SHEN	ST
2.	242UT241HP	LEE JIA YIN	ST
3.			
4.			

Contents

No table of contents entries found.

Task Distribution

No.	Student ID	Student Name	Tasks Completed
1.	242UT244D2	YEAP BOON SHEN	1) Designed the initial UI concept and layout for all dashboard screens (Manager & Employee). 2) Setup frontend (script.js) and backend APIs (PHP) 3) Implemented: Login/Register, Apply Leave, Approval flow, Notifications, Profile, Password Change, Manager Report & User Management 4) Flow chart , ERD
2		Person 2	1) Researched Malaysia Employment Act 1955 to understand leave entitlements. 2) 补齐 and improved Employee-related Figma designs, including: • Login page UI • Employee type selection (Permanent, Contract, Part-Time) UI • Employee level selection Ensured the UI is connected with the backend PHP files. (这个如果你不会加你们可以 code 好了拿到想要的 ui 我帮你们弄) 3) Created the ERD (Entity Relationship Diagram) for the system database.

3	242UT241HP	Lee Jia Yin	1) 补齐 and improved Manager-related screens, including Manager Dashboard, Approvals, User Management, and Reports make sure it works properly. 2) Improved Manager Dashboard with charts and team overview data. UI 3) 在 apply leave 那边 add and create ui 一个可以 import pdf img 等等的 功能和 frame 然后 upload 上去 as a proof Add File Upload Feature (IMPORTANT): • Medical Certificate (MC) • PDF documents • Images (proof) • Preview uploaded files before submit
4		Person D	2) 补齐 and completed Notification, Profile, and Reset/Update Password screens (UI). 3) add forgot passwords (UI) 在 login/signup page 4) Notification UI with read/unread states use two different colour 可能深蓝色代表收到按了就浅蓝色 划走就没有了等等. debugging and ensuring it works correctly

Introduction

Introduce your topic. Tell the reader what the topic is about. Describe the main part and explain the key points of your program.

This project focuses on the design and development of a web-based leave management system called **NexusLeave**. The system is developed for a **hypothetical** company, NexusTech Solutions.

The main purpose of NexusLeave is to replace traditional manual leave management methods with a centralized digital system. The platform allows employees to submit leave applications, check their leave balance, and track request status in real time. At the same time, managers can review and approve requests efficiently.

By using modern web technologies, the system improves communication, reduces errors, and increases overall efficiency in managing employee leave.

Problem Statements

State the **issues/problems** to be **addressed or conditions** to **be improved that lead to the development of the program.**

In many Malaysian small and medium enterprises (SMEs), leave management is still handled manually through paper forms, emails, or messaging platforms such as WhatsApp. This creates several problems.

Firstly, manual calculation of leave balance is prone to human error especially when managing many employees. Secondly, there is a lack of transparency, as employees cannot easily view their leave history or current status.

Objectives

State the objectives of **what you want to achieve with your topic**.

The main objective of this project is to develop a functional prototype that automates the leave management process.

The scope of the system includes:

- A secure login and authentication system
- A user dashboard for employees
- An admin panel for managers
- A leave application and approval system

In addition, the system includes a **Date Calculation Engine** that automatically excludes weekends based on the Kuala Lumpur working calendar. This ensures that only actual working days are deducted from the employee's leave balance.

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The major purpose of this project is to design a functional online employee leave management system that will make the leave application process easy and convenient for employees, as well as automate the process of approving leave applications by managers. The leave management system is intended to take away from employees' hands all manual paperwork in connection with applying for leave and provide the ability to manage their leave applications effectively.

First, the system offers a way for users to log in using their unique credentials to gain access to the portal. After logging in, employees are presented with an interface where they can easily make leave applications and monitor the progress of their application until it is either approved or disapproved. Furthermore, employees are required to upload any supporting documents or images as part of their application for ease of verification purposes.

From the point of view of management, the software comes with an exclusive admin console where managers can conveniently check all the leave applications, process their acceptance or rejection with appropriate comments, and keep track of the attendance status of the entire team. Additionally, the software is designed with an automatic Date Calculation Engine, which automatically filters out weekends according to the Kuala Lumpur office schedule and deducts only working days from the total leave balance of the concerned employees.

Program Scope

Describe the boundaries of the entire program, e.g., features, functions, deliverables, constraints, etc.

Detailed Description

Describe the program in detail which could include but are not limited to:

- A detailed explanation
- System design
- A draft interface design etc.

Screenshot of the program

Provide screenshots of the program with the explanation. It should be treated as a user manual.

2.1 UI/UX Design Strategy in Figma

Before development began, the system was first designed using Figma. This step was important to ensure that the system is user-friendly and visually appealing.

A “Clean Corporate” design style was used, with a blue and white color scheme to represent professionalism and clarity. User flows were carefully planned, including login, dashboard navigation, and leave application processes.

High-fidelity prototypes were created to visualize layout, spacing, and components such as cards, tables, and navigation menus. This helped ensure consistency and better user experience.

2.2 Transition from Design to Development

One of the main challenges was maintaining consistency between the Figma design and the actual implementation.

Vanilla CSS was used to replicate the design details such as spacing, shadows, and layout. This design-to-code approach helped identify potential issues early, such as responsiveness on different screen sizes.

By standardizing UI components, the system maintains a consistent look across all pages.

Technical Architecture

NexusLeave follows a client-server architecture.

- **Frontend:** HTML5, CSS3, JavaScript (ES6)
- **Backend:** PHP 8.x
- **Database:** MySQL

The database includes three main tables:

- users – stores user information and roles
- leave_requests – stores leave applications
- notifications – handles system alerts

This structure ensures efficient data management and system performance.

Reference

State references you refer to complete your program

Appendix

Attach some supporting documents or information that are not included in the previous sections.

You may also include the source code of the created programs. Clear comments are necessary to help the reader understand your code.